

SCSIT

IV Semester

DataBase

Exc. 7

Relational Algebra and SQL for Union, Subtraction and Intersection

Relational Algebra operations

☐ **Basic**

- ☐ *projection*
- ☐ *rename*
- ☐ *selection*
- ☐ Cartesian product
- ☐ **union**
- ☐ **subtraction**

☐ **Derived**

- ☐ **Intersection**
- ☐ *join*
- ☐ division

UNION: $\cup$

Tables: R и S ($R=S$ is allowed)

General Relational Algebra Form: $R \cup S$

Characteristics:

1. Commutativity : $R \cup S = S \cup R$
2. The result contains tuples from both table, but all duplicates are removed.
3. Tables must be **COMPATIBLE**(???)

Compatibility

1. Both table must have same **degree** (Number of columns)
2. Appropriate table attributes (one from the first table, and one from the second one) must be defined on **SAME DATA DOMAINS**
3. If the appropriate table attributes are with different names, it is wise to make ***renaming***.

Ex. 1:

Compatibility or not

T1=

A	B
1	A
3	Б
3	B
<i>null</i>	Г

T2=

A
4
5
6
6
8

No compatibility

T1=

A
4
5
6
6
8

T2=

A
A
B
C
D

No compatibility

Ex. 2:

Compatibility or not

T1=

A	B
1	A
3	B
2	T

T2=

C	D
O	7
R	5
T	2
Q	1

T3=

A
S
T
Q
A
C

T4=

A
A
C
D
W

YES (*Why?*)

$A \equiv D; B \equiv C$

YES
(*why?*)

Lets make $T1 \cup T2$ ($A \equiv D$; $B \equiv C$)

1. First, rename the columns (only in the result pane)

- You can make several types of renaming: second table columns get names from the first table appropriate columns, or first table columns get names from the second table appropriate columns, or both tables appropriate column get new result name

2. Then, make the Union

$T1 \cup \rho_{(D/A, C/B)} T2$ or

$\rho_{(A/R, B/S)} T1 \cup \rho_{(D/R, C/S)} T2$

(T2 renaming process)

(T1 and T2 renaming process)

And the result is...?

Union Result

T1 $\cup$ $\rho(D/A, C/B)$ **T2**

A	B
1	A
3	B
2	T

$\cup$

C	D
O	7
R	5
T	2
Q	1

=

A	B
1	A
3	B
2	T
7	O
5	R
1	Q

Duplicate

T3 $\cup$ **T4** = ? Any volunteers 😊⁸

UNION SQL

```
( SELECT A1, A2,..An  
  FROM table1  
  [WHERE condition; Group by...]      )
```

UNION

```
( SELECT B1 as A1. B2 as A2,..Bn as An  
  FROM table2  
  [WHERE condition; Group by...]      )
```

```
[order by, group by ...]
```

Tables:

work_place and pensioner

ID_PLACE	TITLE	SALARY	SECTOR	PHONE	VEHICLE	LAPTOP
1	MANAGER	33500	MARKETING	1	1	1
2	DBA	27000	IT	0	0	1
3	ANALYST	28500	IT	1	0	1
4	PROGRAMMER	25000	IT	0	0	1
5	SECRETARY	12500	ADMINISTRATION	0	0	0
6	DOORMAN	7000	ADMINISTRATION	0	0	0
7	ACCOUNTANT	10500	FINANCE	1	0	0
8	TREASURER	11500	FINANCE	1	0	0
9	DRIVER	7500	SERVICE	1	1	0
10	Event manager	21500	MARKEING	1	1	1

ID_PENSIONER	NAME_P	PENSION
1	pERO sTOJANOVSKI	11500
2	Vaska Markovska	8700
3	Dr. Meto Jovanov	23000
4	Blagoja Spirov	12500
5	Petar Sokolov	14680
6	Milena vikova	7300
7	Dragica Pavlova	12000

Ex 3.

Find union of monthly salaries/pensions and employee/pensioner names

(select Title as Name, salary as INCOME
from work_place)

UNION

(select name_p as Name,
pension as INCOME
from pensioner)

Name	INCOME
ACCOUNTANT	10500.0
ANALYST	28500.0
Blagoja Spirov	12500.0
DBA	27000.0
DOORMAN	7000.0
Dr. Meto Jovanov	23000.0
Dragica Pavlova	12000.0
DRIVER	7500.0
Event manager	21500.0
MANAGER	33500.0
Milena vikova	7300.0
pERO sTOJANOVSKI	11500.0
Petar Sokolov	14680.0
PROGRAMMER	25000.0
SECRETARY	12500.0
TREASURER	11500.0
Vaska Markovska	8700.0

Sort the result set according the *income* value

Ex 4. Find union of pensioners monthly incomes and names with salaries, titles and sectors for the employers in work_place

(select Title as Name,
salary as INCOME,
sector
from work_place)

UNION

(select name_p as Name,
pension as INCOME
from pensioner)

**These SQL
Statements are
NOT Compatible**

Server: Msg 205, Level 16, State 1, Line 30

All queries in an SQL statement containing a UNION operator must have an equal number of expressions in their target lists.

1. Find salaries above 25000 and pensions below 10000. Result column should be named as EXTREMES

2. Find salaries and pensions between 10000 and 25000. Rename the column into "Normal_income".
(notice what happened with duplicates)

3. Find the separate sums of salaries and pensions.

4. Expand Result set from (3) with additional column with fixed data "SALARY" and "PENSION". Column name should be ITEM

SUM_INCOME	ITEM
89680.0	PENSION
184500.0	SALARY

AMOUNT	UNIT
9750.0	ADMINISTRATION
11000.0	FINANCE
26833.0	IT
21500.0	MARKEING
33500.0	MARKETING
89680.0	PENSION
7500.0	SERVICE

5. Find average salaries per sector and sector. place them in same result set with average pensions. Columns should be named as UNIT and AMOUNT.

6. Find sum of salaries in IT sector and sum of salaries below 10000.

AMOUNT	UNIT
16000.0	PENSION
80500.0	IT

**7. Find average salaries for IT and Administration.
List the Amount and Sector name.**

8. Find pension minimum, maximum, sum and average of pensions. All data should be placed in one column named as DATA_PROCESSING, and in second column name as ITEM should be names like: Minimum pension, Max pension etc.

Subtraction -

Tables R and S ($R=S$) is allowed)

R - S

Characteristics:

1. NON - Commutativity : $R - S \neq S - R$
2. The result set contains touples that belong to table R, but don't belong to S
3. Tables **must be compatible**, at least in the “don't belong” condition part

Ex. 5.

r

A	B
a	1
a	2
b	1

s

A	B
a	2
b	3

r - s

A	B
a	1
b	1

What about
S-R?

SQL for Subtraction

Ex. We want to find data that are part of table1 but they have to fulfilled the condition that data from column A1 are not part of Column b1 which is part of Table2

SELECT A1, A2,..An

FROM *table1*

WHERE [condition ... and] A1 **NOT IN**
(**SELECT** B1 **FROM** *table2*
[**WHERE** condition...])

Note: A1 may or may not be part of the Result set

Ex 6.

□ Find and sort all salaries that are bigger than the minimal

SELECT salary

FROM work_place

Where salary NOT IN

(select min(salary) from work_place)

order by salary

9. Find all salaries (amount and work place title) that are not amounts of pensions

10. Find all pensions (amount and pens. Name) that are not salary amount

11. Find all pensions and p. names for those pensions that are neither minimal nor maximal.

12. Find all salaries that are bigger than the 2 smallest salaries

INTERSECTION $\cap$

Tables R and S (also, $R=S$ is allowed)

$$R \cap S = R - (R - S)$$

Characteristics:

1. Commutativity: $R \cap S = S \cap R$
2. The result contains tuples that belong to R and S in the same time
3. **Tables must have at least one compatible part (column set)**

Ex. 7.

r

A	B
a	1
a	2
b	1

s

A	B
a	2
b	3

R $\cap$ **S** =

A	B
a	2

INTERSECTION SQL

1. We need to find values from A1 in B1

```
SELECT A1  
FROM table1  
WHERE [condition ... And] A1 IN  
(SELECT B1 FROM table2 [WHERE condition...])
```

2. We create equalization of A1 and B1 values

```
SELECT A1 FROM table1, table2  
WHERE table1.A1 = table2.B1 [condition]
```

Note: A1 may or may not be part of the Result set

plata
12500.0
11500.0

naziv	sektor	plata
Секретар	Администрација	12500.0
Благајник	финансии	11500.0

13. Find all salaries that have same values as some pensions

```
select salary
from work_place
Where salary in (select pension from pensioner)
```

14. Find salary amounts, titles and sectors for salaries that have same values as some pensions.

Can you create SQL statements with equalization?

One more SQL 😊

15. Find salary amounts, titles and sectors for salaries that have same values as some pensions. **But also find Pensioner names as additional column in the result set**

- Can you make this with "IN" operator ?

	amount	title	sector	name_p
1	12500.0	SECRETARY	ADMINISTRATION	Blagoja Spirov
2	11500.0	TREASURER	FINANCE	pERO sTOJANOVSKI